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RESEARCH ARTICLE

The Path to an Inclusive Identity: A Non-Authoritative Approach

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ABSTRACT The goal of providing legal identity to all individuals remains a global challenge. Current digital identity models depend on the existence of authoritative issuers, which are not always available, particularly in Low-Income Countries (LICs) and Lower-Middle-Income Countries (LMICs). This limitation severely restricts the coverage of identity systems. In response, new approaches such as the non-authoritative identity model have emerged, enabling entities with local reputation to serve as primary sources of information about individuals. However, existing literature does not yet offer a comprehensive solution for implementing such systems. In this paper, we present Nexus, a data fusion module that enables reputable entities to serve as primary providers of identity information, facilitating the design and implementation of digital identity systems that complement the State or other centralized authorities. A central element of the proposed data fusion component is a novel scheme combining truth discovery algorithms and reputation tracking. The feasibility, behavior, and performance of the proposed approach are demonstrated through experiments conducted on multiple datasets. In addition, Nexus was evaluated using a dataset from a health line service in a LIC, thereby demonstrating its applicability in real-world settings. This work represents an essential step toward eliminating barriers to pursuing a comprehensive legal identity.

INDEX TERMS Data fusion, digital identity, non-authoritative identity, reputation tracking, truth discovery.

I. INTRODUCTION

According to data published by The World Bank, globally, around 850 million people lack an official proof of identity [1]. This reality affects 1 in 9 people, impacting, in particular, Low-Income Countries (LICs) and Lower-Middle Income Countries (LMICs). Over 90% of the affected people live in LICs and LMICs, and nearly half of them are children [1]. This global-scale problem limits access to essential services such as healthcare, education, and social security, while also restricting participation in economic and political processes.

The United Nations (UN) has long recognized legal identity as a fundamental right, having established in Article 6 of the Universal Declaration of Human Rights [2], and in

Article 16 of the International Covenant on Civil and Political Rights [3] that every individual has the right to be recognized everywhere as a person before the law. These commitments are reflected in the UN Sustainable Development Goals (SDGs) outlined in the 2030 Agenda [4], specifically Goal 16.9, which aims to provide “legal identity for all”.

Nevertheless, several challenges hinder the efforts to provide legal identity to everyone. Ensuring equitable access to legal identity across all segments of the population remains a difficult task, particularly given the infrastructural and financial demands of issuing identity documents, barriers that are especially acute in LICs and LMICs [5], [6], [7]. The enrollment or registration of individuals in a digital identity system is the first step in the process of establishing a legal identity [8]. However, this process often encounters significant obstacles. For instance, long distances to the nearest registration points, cultural and language barriers,

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registrants' lack of trust in the process, and associated costs are some of the challenges that can hinder the enrollment of individuals [8], [9]. The existing obstacles in this process are a key factor in accentuating inequality and exclusion, severely limiting the coverage of both new and existing identity systems.

The Self-Sovereign Identity (SSI) model appeared as an alternative to traditional centralized and federated identity models existent in the majority of countries, shifting control of user data from centralized entities (*e.g.*, official civil registries) to individuals themselves [10], [11], [12]. By allowing individuals to fully own, control, and manage their identity information without relying on any central authority, SSI improves on aspects such as privacy, security, and interoperability [10].

Current digital identity models, such as SSI, rely on the existence of a trusted issuer to provide accurate information about individuals. An issuer may be an authoritative entity, defined as an official entity or a third-party acting on behalf of the State, or it can be an individual asserting their own information [10]. There are various scenarios where an authoritative entity is not available, particularly in LICs and LMICs. Moreover, when information is provided by individuals, there is no guarantee regarding its veracity, as individuals may benefit from providing false information. This severely limits the applicability of SSI systems, failing to provide an answer to the challenge of providing an all-encompassing legal identity.

In response to this issue, the non-authoritative identity model has emerged as a new approach [13], which allows locally established entities to act as the primary sources of information about individuals. Early work in this area focuses on the concept and requirements of a non-authoritative identity system, addressing the challenges associated with leveraging social trust relationships to establish a digital identity. However, existing literature does not provide a practical solution for the implementation of such systems, particularly in terms of consolidating multi-source data characterized as incomplete, divergent, or duplicate.

In this paper, we address this gap by presenting *Nexus*, a data fusion module that enables the enrollment of individuals in non-authoritative identity systems. By allowing reputable entities to act as primary sources of information about people they interact with, we enable the design and implementation of a digital identity system capable of complementing the State or other central authorities. *Nexus* is capable of consolidating multi-source information and producing a single accurate representation for each individual. Moreover, the proposed solution tracks the contribution history of each source, allowing it to evaluate the reliability of sources and use this information to improve its accuracy over time. A publicly available implementation of *Nexus* is provided¹ as a core contribution of this

work, facilitating reproducibility and future research in non-authoritative identity systems.

To demonstrate the real-world applicability of *Nexus*, an evaluation was conducted using data from Guinea-Bissau's 24-hour Health Line service. This dataset comprises information on more than 5,000 individuals collected by line operators during telephone interactions. The resulting analysis demonstrates the feasibility of applying *Nexus* in real-world settings, highlighting its potential both to enhance the enrollment process in digital identity systems, particularly in contexts where authoritative entities are unavailable or have limited capacity, and to improve service delivery, resource allocation, and policy-making across multiple sectors, including healthcare, social services, and humanitarian aid.

Nexus aims to be the first step towards an inclusive digital identity system that can be implemented, in particular, in LICs and LMICs, where the lack of an official issuer represents a significant barrier to providing legal identity to everyone. It allows confidence in attributes about individuals to incrementally increase until a predefined threshold is reached, at which point these attributes may be utilized within authoritative or official systems, thus presenting a key advancement in the pursuit of providing legal identity to everyone.

The remainder of this paper is structured as follows. Section II provides background on digital identity models, non-authoritative identity systems, and data fusion techniques, positioning *Nexus* within the existing landscape of digital identity. Section III presents the design goals, architecture, and detailed components of *Nexus*, including the truth discovery module and reputation module, along with a mathematical analysis of adversarial scenarios. Section IV outlines the evaluation methodology for the proof of concept and the datasets employed to assess our proposed solution. Section V presents comprehensive results on accuracy and parameter sensitivity across synthetic and curated datasets, whilst showcasing the real-world applicability of *Nexus* using data from Guinea-Bissau's 24-hour Health Line service. Finally, Section VI discusses the findings, implications, and identifies directions for future work.

II. BACKGROUND AND RELATED WORK

The process of establishing and subsequently using an individual's digital identity comprises multiple stages, often referred to as the identity lifecycle [5], [14]. The first stage consists of the registration or enrollment of individuals in a digital identity system, which is the main focus of this work. Enrollment involves the identity claim, where individuals claim their identities by providing personal data and supporting evidence, and identity proofing, where the authenticity and accuracy of the provided information are verified [5].

After the registration stage, credentials (*e.g.*, cards or certificates) are issued to the individual. The third stage consists of using these credentials as proof of identity when accessing services (*i.e.*, their daily use). The last

¹Implementation available at <https://github.com/dissect-project/nexus>

stage of the lifecycle involves managing identity data and credentials [14]. This stage has a dynamic nature, involving the update and re-verification of identity attributes that change or evolve over time (*e.g.*, address or marital status), as well as the retirement or revocation of credentials upon discovery of fraud or upon an individual's death.

In recent years, new decentralized models, such as Self-Sovereign Identity (SSI) [10], [12], [15], [16], [17], have emerged as an alternative to traditional centralized identity models. SSI presents a shift in control, allowing individuals and organizations to manage their own identities without the need for a central authority [11]. This approach empowers users to have better control over their personal information, enhancing privacy and security [10]. SSI has been identified as a promising approach in humanitarian contexts, particularly for refugees [18], [19]. An example of a SSI deployment in a LIC is reported in [20], where Kiva and the Government of Sierra Leone collaborated to provide financial services to individuals without access to traditional banking. However, the adoption of SSI systems by governments and organizations is not without its challenges. Lack of standard solutions, technical complexity, and early-stage ecosystem development are among the main obstacles for SSI adoption in legal contexts [12], [21].

Verifiable Credentials (VCs) are a core component of the SSI model, enabling users (*i.e.*, holders) to share their identity information in a secure and privacy-preserving manner [10], [12], [22]. The *issuer* of a credential may be an authoritative entity (such as governmental agencies or universities), or it can be an individual asserting their own information. An authoritative entity consists of an official source which is widely recognized to have legal or regulatory authority over the information it provides (*e.g.*, civil registry), or it can also be a third-party entity acting on behalf of the State (*e.g.*, a notary) [13]. Credentials include various attributes related to the individual they were issued to (such as name, date of birth, or address), as well as a digital signature from the issuer, which guarantees the authenticity and integrity of the credential. A *verifier* requests proofs from holders on attributes from one or more VCs. The relationship between issuers, holders, and verifiers is often referred to in literature as the SSI trust triangle [10], as shown in Figure 1.

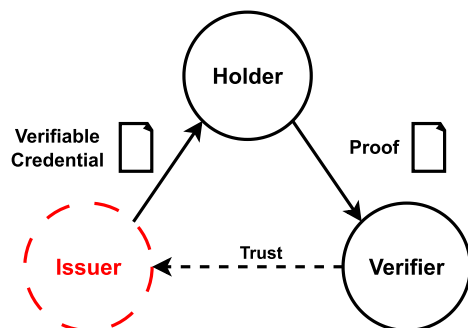


FIGURE 1. Trust triangle in the SSI ecosystem.

An important part of the trust triangle resides in the existing trust relationships, specifically, the relationship between the verifier and the issuer. The verifier must trust the issuer to be authoritative for the holder's attributes that are contained in a proof that it receives, given that each verifier determines its own trust rules. A verifier may trust a university to be authoritative for the holder's educational qualifications, but may not trust the same university to be authoritative for the holder's date of birth. In order to assist verifiers in determining which credentials from which issuers they should trust, a Verifiable Data Registry (VDR) maintained by a Governance Authority is used, containing a list of issuers and the properties they are considered authoritative for [10]. However, a key challenge in this model resides in cases where the required trust relationships cannot be established.

There are various scenarios where authoritative entities are either unavailable or have a limited scope, particularly in LICs and LMICs. Remote locations, cultural or linguistic barriers, and the lack of trust in the process are some of the challenges in the enrollment of individuals in a digital identity system and subsequent issuance of credentials [8]. Moreover, relying on individuals to provide information about themselves also presents its challenges, as there is no guarantee that an individual does not provide false information with the intention of benefiting from it. In these cases, it becomes impossible to establish a trust relationship between verifiers and issuers. Such obstacles severely hinder the applicability of SSI systems in LICs and LMICs, leaving a critical gap in efforts to complement the State in providing legal identity to all segments of the population.

A. NON-AUTHORITATIVE IDENTITY

As a response to this limitation, the concept of non-authoritative identity has emerged as an extension to the SSI model that does not depend on the implicit trust relationship between issuer and verifier [12], [13], [23]. In a non-authoritative approach, instead of being issued by a central authority, credentials are composed of attributes supplied by various sources [13]. These sources can be non-governmental organizations cooperating with local leaders to complement or replace public administration in delivering basic services such as healthcare or education. In many cases, these entities have a better understanding of the population and are locally well-recognized as reputable structures [24]. Moreover, these organizations often possess more extensive and more advanced technological resources than the State itself [5]. As a result of their deep integration with local realities, these entities are capable of acting as sources of information with the potential to enable the design and implementation of a digital identity system striving to be inclusive and complementary to the State.

Figure 2 illustrates how a non-authoritative model enables the fostering of existing social trust relationships. In this model, the *issuer* is replaced by multiple reputable entities (*i.e.*, sources), such as healthcare providers or local

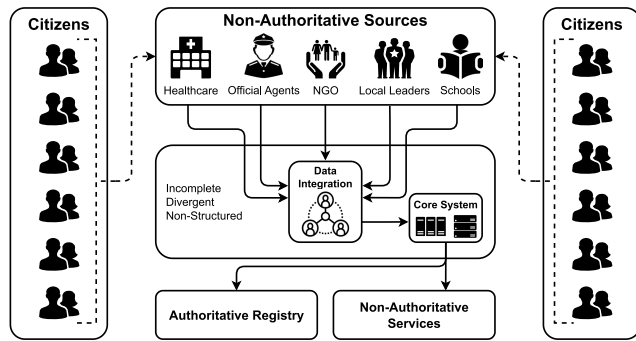


FIGURE 2. Overview of a non-authoritative architecture [13].

leaders who can provide information (*i.e.*, claims) about individuals they interact with. Information provided by these sources is expected to be incomplete, unstructured, or even divergent, requiring the system to gradually consolidate the information to create a single accurate representation for each individual. Confidence in information is directly linked to the assessed trustworthiness of its source and evolves through corroboration by multiple independent sources. This corroborated information, in turn, reinforces trust in the original source. The resulting representations can then be used to issue credentials that may be employed in either non-authoritative services or traditional authoritative services, thereby acquiring legal value. Achieving this goal requires two core components within the non-authoritative identity model: (i) a data integration process that consolidates information from multiple sources, and (ii) a reputation system that continuously assesses and updates the reliability of those sources over time.

B. DATA FUSION

The data integration process plays a crucial role in combining information from multiple sources into a single, consistent, and accurate representation, which can then be employed in a non-authoritative scenario. Despite not being specifically designed for identity systems, existing literature identifies three essential steps in the data integration process [25], [26]: (i) schema mapping, (ii) data fusion, and (iii) duplicate detection.

Data fusion, also referenced in literature as data consolidation, encompasses the process of merging information from different sources into a single coherent representation [25], [27], [28]. This process is essential to ensure that raw data originating from various sources representing the same real-world object can be merged (or fused) into a concise and coherent representation [25], [28], [29].

In some instances, information provided by different sources may be contradictory, requiring the system to resolve these conflicts and produce a unified representation. This task is especially relevant for the duplicate detection process, as conflicting information increases its difficulty. Truth discovery is a subtask within data fusion that focuses on this

issue, evaluating source reliability and data consistency to determine which value is most likely to correspond to an accurate representation of reality [30], [31]. The literature commonly classifies data processed by truth discovery algorithms into three types: categorical, string, and continuous [27]. Additionally, heterogeneous data is referenced in literature as values encompassing any combination of these three data types [30].

Some of the most well-known truth discovery algorithms include CRH [32], 2-Estimates [33], 3-Estimates [33], Investment [34], LTM [35], GTM [36], and TruthFinder [31]. These algorithms introduce different approaches to the challenge of inferring the truth from multi-source conflicting data. One key difference between these algorithms lies in their support regarding the types of input data. From the previously presented algorithms, only TruthFinder and CRH are capable of handling heterogeneous data types [37], a reality of identity systems.

TruthFinder employs Bayesian analysis to iteratively estimate source reliability and identify values viewed as the truth [31], [37]. To support heterogeneous data, different implication functions are employed for each data type [31]. The implication between values measures how much the confidence in a value should be affected based on the confidence score of a second value. It is captured through similarity functions tailored for each data type, returning a score in the $[-1, 1]$ range, where 1 represents maximum similarity between two values, and -1 represents the opposite.

CRH infers truths from conflicting sources by formulating the issue as an optimization problem to minimize the overall weighted deviation between the input and the values identified as truths. Heterogeneous data is supported by using distinct loss functions for each data type [32].

In both of these iterative algorithms, an estimation of source reliability is produced at the end of execution. However, this estimation is not preserved for future runs, resulting in the loss of source reliability information between executions and forcing the algorithms to assign uniform weights to all sources at the start of each new execution. This limitation, known as the source reliability initialization problem [37], remains an open issue in the literature [30].

More recent work [38], [39], [40] has introduced dynamic truth discovery methods capable of handling scenarios where information arrives sequentially rather than in batches. By continuously updating both the inferred truth and source reliability, these algorithms can adapt to evolving data over time. However, a key limitation of these approaches lies in their inability to accommodate heterogeneous data [38], [39], [40]. Furthermore, the methods proposed in [38] and [40] do not retain historical data, preventing prior executions from being revisited when sources are later found to have provided inaccurate information.

Another approach to data fusion relies on machine learning-based techniques [29], [41], [42], [43]. These methods typically require labeled data and substantial amounts

of training data to achieve effective performance levels [44]. A major challenge, however, is that ground truth data is often unavailable in real-world settings. Moreover, even when training data exists (e.g., the TruthSeeker dataset [45]), it is difficult to ensure that it adequately captures the full range of variations encountered in practice. Such variations may include differences in pronunciations, spellings, and formatting of names, addresses, or other attributes, which can vary significantly across regions, cultures, dialects, and languages. In addition, constructing and using training datasets that contain personal information raises significant privacy concerns under regulatory frameworks like the General Data Protection Regulation [46] and the EU Artificial Intelligence Act [47], which mandate strict data governance and risk management. Collectively, these challenges hinder the creation of comprehensive and representative training datasets, thereby limiting the robustness and real-world applicability of machine learning-based data fusion approaches in the context of digital identity.

III. NEXUS

In this section, we present Nexus, a data fusion solution designed to support a non-authoritative identity model by consolidating information from multiple sources into a single, accurate representation for each individual. From this point forward, assertions made about an individual will be referred to as *claims* regarding that individual. In a non-authoritative scenario, incoming data is expected to be heterogeneous, duplicate, and often conflicting, as it originates from various sources that may possess differing degrees of knowledge about an individual.

Sources exhibit varying levels of reliability, which may also change over time (e.g., a source may be discovered to provide false information). A key differentiating aspect of the approach used in Nexus is its ability to track each source's contribution history by evaluating the quality of the claims it provides. This functionality is particularly important for monitoring the reliability (or reputation) of sources, which determines the influence of their contributions. Consequently, Nexus can use this knowledge to improve the accuracy of the data fusion process over time and to revisit and correct previous executions upon discovering sources that provide low-quality information.

A. DESIGN GOALS

The primary objective of this work is to present a data fusion solution tailored for non-authoritative environments, capable of consolidating claims from multiple sources into a single, accurate representation for each individual. Nexus aims to improve the enrollment process in cases where official entities are unable to operate (e.g., in remote communities). To achieve this objective, and in consideration of the challenges previously identified, we define the following design goals:

- **Design Goal 1:** Heterogeneous data support: the solution should support heterogeneous data types often found in identity records, including categorical, string, and continuous data.
- **Design Goal 2:** Divergent data consolidation: the solution should be capable of handling divergent data, being able to determine which values are more likely to correspond to the truth.
- **Design Goal 3:** Confidence tracking: the solution should be able to provide a confidence score for each record in the system, thus allowing the source providing that value to be evaluated.
- **Design Goal 4:** Reputation tracking: the solution should be able to track and update the reputation of sources over time, using this information to improve the accuracy of the information.
- **Design Goal 5:** Unreliable source handling: the solution should be capable of revisiting parts of the data fusion process upon the detection of an unreliable source.

B. ARCHITECTURE

The proposed architecture for Nexus, illustrated in Figure 3, demonstrates how our solution is capable of receiving a set of claims and outputting the one most likely to correspond to the truth (Design Goal 2), as well as the confidence scores for each supplied claim (Design Goal 3) and the respective reputation scores for all sources (Design Goal 4). There are 3 modules inside the data fusion component: the truth discovery module, the reputation module, and the evaluation module. The interaction between these modules, as well as the information exchanged, is shown in Figure 3.

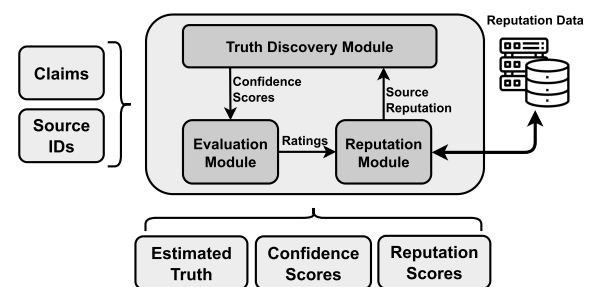


FIGURE 3. Architecture of Nexus.

Upon receiving the required input (claims and their respective source identifiers), the truth discovery module requests the reputation scores for each participating source from the reputation module. If a source submits its first claim, the reputation score is initialized with a uniform value. If it already has previous contributions, the respective reputation score is fetched from the reputation module's database.

Using and tracking source reputation is a key aspect differentiating our approach from existing truth discovery algorithms. This allows us to perform a single iteration of the truth discovery process, as opposed to multiple iterations required by other algorithms, since we are not

TABLE 1. Frequently used notations.

Notation	Description
i	Individual i
s	Source s providing claims about individuals
c	Claim c provided by sources
i_c	Represents the individual i that claim c is about
r_s	Reputation score of source s provided by the reputation module
$\mathcal{S}_c$	Represents the set of sources that have provided a claim c
$\omega(c)$	Weight of claim c
$\text{imp}(c' \rightarrow c)$	Quantifies the implication (<i>i.e.</i> , similarity or dissimilarity) between claims c' and c
$\sigma(c)$	Quantifies the influence of the remaining claims about an individual i_c
ρ	Controls the weight of $\sigma(c)$ (influence between related claims) in the consolidation process
γ	Dampening factor that limits the rate at which confidence scores evolve
$\phi(c)$	Confidence score of claim c , <i>i.e.</i> , the probability of claim c being true
$\vec{r}_s$	Binary rating vector for a source s
k	Number of levels present in the rating vector $\vec{r}_s$
$\vec{a}$	Default base rate in the reputation system
C	A-priori constant which influences the impact of new ratings
α	Total evidence for a source (sum of ratings of a source and the base rate)
λ	Longevity factor of existing ratings for a source (<i>i.e.</i> , controls the influence of past ratings)
$\vec{R}_s$	Aggregate ratings vector of a source (<i>i.e.</i> , sum of all ratings for a source)
$\vec{S}$	Multinomial probability vector for a source s with k levels

required to update the trustworthiness of sources in runtime. Furthermore, by retaining historical data from previous executions of the truth discovery process, Nexus is able to revisit and re-evaluate earlier inferences in light of updated source reputations. This functionality is particularly relevant in scenarios where sources initially perceived as reliable are later found to have provided false or misleading information.

After estimating the confidence scores for each claim, the truth discovery module forwards this information to the evaluation module, which maps the respective confidence scores into ratings for each source. These ratings are then forwarded to the reputation module, which updates and stores the reputation scores of each source accordingly, thus ending what we designate as an iteration inside Nexus. A detailed description of the operation of each module is provided in Section III-C.

Regarding the task of duplicate detection, Nexus assumes that the input data has already been processed to ensure that each real-world entity is represented only once in the system. This assumption is based on the fact that duplicate detection is a complex task that often requires domain-specific knowledge and techniques, which are outside the scope of this work. Some approaches, such as the use of pre-trained language models [48], [49], [50], have shown promising results in this area by demonstrating both their versatility and high-performance.

C. COMPONENTS AND INTERACTION

The architecture presented in Figure 3 is composed of two main components: the truth discovery module and the reputation module. The truth discovery module estimates the truth from heterogeneous data, while the reputation

module tracks and updates the trustworthiness of sources participating in the process over time. Additionally, the source evaluation module translates the contributions of each source into ratings that are subsequently utilized by the reputation module. The notation adopted in this work is summarized in Table 1 and further described in the following section, along with a detailed description of the components comprising Nexus and their underlying interactions.

1) TRUTH DISCOVERY MODULE

To implement the truth discovery module, we rely on truth discovery algorithms capable of handling heterogeneous data. From the algorithms presented in Section II, only TruthFinder and CRH support heterogeneous data [37] (Design Goal 1). A key difference between these algorithms lies in their output. TruthFinder outputs a confidence score in the range $[0, 1]$ for each value, whilst CRH outputs a source weight in the range $[0, +\infty]$ for each participating source [31], [32]. Considering Design Goal 3, we require the truth discovery algorithm to output a well-defined confidence score for each value. The confidence score, or fact confidence, represents the probability of a claim being true or not. This information is crucial for understanding the degree of trust we can place in the estimated truth, as well as to assess the contribution of each source. Therefore, TruthFinder represents the alternative that best suits the design goals of our solution.

Despite their effectiveness in addressing the truth discovery challenge, existing algorithms have several key limitations. One of the most significant issues is the source reliability initialization problem [30]. Algorithms such as TruthFinder, for example, assign uniform reputation scores

to all sources at the beginning of each execution. Source trustworthiness and fact confidence are then iteratively derived from one another, either for a predetermined number of iterations or until the change between iterations falls below a given threshold [31]. Furthermore, by assigning uniform reputation scores, existing approaches overlook relevant information that could be used to track the historical performance of sources.

We diverge from the methodology of Yin et al. [31] by presenting each source's reputation score at the outset of the truth discovery process. This approach eliminates the need for multiple successive iterations, since only the confidence score for each claim is required to be calculated. To determine which claims are most likely to represent the truth, our implementation relies on the following pieces of information: (i) a source identifier for each claim, (ii) the claims provided by each source, (iii) the individuals to which each claim pertains, and (iv) the data type of the claims.

$$\omega(c) = \sum_{s \in \mathcal{S}_c} -\ln(1 - r_s) \quad (1)$$

$$\sigma(c) = \sum_{i_{c'}=i_c} \omega(c') \cdot \text{imp}(c' \rightarrow c) \quad (2)$$

$$\phi(c) = \frac{1}{1 + e^{-\gamma \cdot (\omega(c) + \rho \cdot \sigma(c))}} \quad (3)$$

The first step of this process consists of calculating the weight ω for each claim using (1). The weight of a claim $\omega(c)$ is a direct result of the reputation (or trustworthiness) of the sources providing that exact claim. The set of sources providing a claim c is represented by $\mathcal{S}_c$. However, the weight alone does not take into account the existence of different claims about the same individual and the relationship (or implication) between them. The implication from claims c' to c , represented by $\text{imp}(c' \rightarrow c)$, is a metric that allows us to quantify how similar (i.e., high implication) or how contradicting (i.e., low implication) they are. For categorical data, direct comparison is used to assess similarity. Equal values return 1, while contradicting ones return -1. To capture the implication between string data, we use the Jaro-Winkler algorithm [51], [52] since it is capable of addressing issues such as typos and phonetic variations when measuring similarity. Implication between continuous data is calculated using Euclidean distance. For all data types, the results are normalized in the interval $[-1, 1]$. A relevant aspect of our implementation is the possibility of updating these implication functions or adding new ones to support additional data types.

The influence of the remaining claims about an individual i_c , represented by $\sigma(c)$, is calculated with (2). By incorporating implication between claims into the proposed solution, we can capture the relationship between them. In the cases where $\sigma(c) > 0$, the remaining claims about the same individual tend to support c . Conversely, when $\sigma(c) < 0$, it suggests that the remaining claims tend to contradict c .

Algorithm 1 Pseudocode for the Truth Discovery Module

```

1: Input:  $[(s_j, c_j, i_{c_j}, \text{data type}, r_{s_j}) : j = 1, \dots, m]$ 
2: Output:  $[(c_j, \phi(c_j)) : j = 1, \dots, m]$ 
3:
4: //Step 1: Calculate the weight of each claim
5: for each claim  $c$  do
6:    $\omega(c) = \sum_{s \in \mathcal{S}_c} -\ln(1 - r_s)$  //(Equation (1))
7: end for
8:
9: //Step 2: Calculate the influence of the remaining claims
   about the same individual
10: for each claim  $c$  do
11:    $\sigma(c) = \sum_{i_{c'}=i_c} \omega(c') \cdot \text{imp}(c' \rightarrow c)$  //(Equation (2))
12: end for
13:
14: //Step 3: Calculate the confidence score for each claim
15: for each claim  $c$  do
16:    $\phi(c) = \frac{1}{1 + e^{-\gamma \cdot (\omega(c) + \rho \cdot \sigma(c))}}$  //(Equation (3))
17: end for
18:
19: return  $\phi(c)$ 
  
```

However, several challenges arise when calculating the confidence score of c using only $\omega(c)$ and $\sigma(c)$. The first one resides in source dependence, which can lead to overly high confidence scores (e.g., due to copying). This challenge is solved with the inclusion of a dampening factor $\gamma \in [0, 1]$, which compensates for the problem of overly high confidence scores by limiting the rate at which scores evolve. The second challenge lies in controlling the weight of related claims about the same individual when calculating the confidence score. This is addressed by including a parameter $\rho \in [0, 1]$ to control the influence between related claims. A value of $\rho = 0$ removes $\sigma(c)$ from the confidence score entirely, while a value of $\rho = 1$ maximizes the impact of $\sigma(c)$ in the final result. Lastly, the possibility of having negative confidence scores in cases where a claim c conflicts with the information provided by sources with higher reputation scores has to be addressed. To solve this, the confidence score ϕ results in a function with similar behavior to the Sigmoid function [53], as shown in (3). By using a function with this behavior, the resulting confidence score is guaranteed to always be between 0 and 1, even in cases where the result of $\gamma \cdot (\omega(c) + \rho \cdot \sigma(c))$ is outside this interval by a significant margin.

The data flow of the truth discovery module is presented in Algorithm 1. The input of the algorithm consists of m claims provided by sources, along with the respective source identifiers, the individuals to which the claims pertain, the data type of each claim, and the reputation score of each source. The output consists of a list of tuples containing each claim and its respective confidence score $\phi(c_j)$.

2) EVALUATION MODULE

The evaluation of a source's contribution, which we refer to as source rating $\vec{r}_s$, is a direct product of the resulting confidence score ϕ of claims provided by sources. A source rating $\vec{r}_s$ is represented by a vector containing k elements (or levels) with a binary state space, being that each source rating is only allowed to have a single level set to 1. The higher the index of the level set to 1, the better the source's rating. The index for the level to be set to 1 is calculated following (4).

$$\vec{r}_s = (r_1, r_2, \dots, r_k), r_{index} \in \{0, 1\}, index = 1 \dots k$$

$$index = \begin{cases} \lfloor \phi(c) * k \rfloor & \text{if } \phi(c) \in [0, 1[\\ k & \text{if } \phi(c) = 1 \end{cases} \quad (4)$$

For an index to be mapped to a position in the source rating vector, it is required to be an integer. The *floor* function is employed to drop the decimal part of the product between k and the confidence score. Hence, the result of (4) will always be an integer value (e.g., $\lfloor 5.4 \rfloor = 5$).

3) REPUTATION MODULE

This module relies on reputation systems for receiving source ratings, calculating their respective reputation scores, and storing all the related data (Design Goal 4). By mapping the contributions of sources into a binary state space through a Bayesian system, we can calculate reputation scores by statistically updating beta probability functions [54]. Given that the size of this state space can exceed 2, the reputation system follows a Dirichlet distribution.

The Dirichlet distribution [55] describes the probability distribution over a k -component random variable $p(\theta_i)$, $i = 1 \dots k$ with sample space $[0, 1]^k$ and subjected to the simple additivity requirement $\sum_{i=1}^k p(\theta_i) = 1$. The total evidence $\alpha(\theta_i)$ for each singleton event θ_i represents the confidence about the likelihood of θ_i , being expressed by (5).

$$\alpha(\theta_i) = r(\theta_i) + Ca_k \quad (5)$$

Vector $\vec{r}$ represents the *a posteriori* evidence, that is, the ratings that are supplied to the reputation module. The *a priori* base rate is represented by a_k , and in case no other evidence is available, it is used to determine the probability distribution of events. The base rate for uniform distributions is given by the expression $a_k = 1/k$. As more evidence becomes available, the influence of the base rate is reduced until the evidence alone determines the probability distribution of the events. C represents the *a priori* constant, which can be set to 2 when a uniform distribution over binary state spaces is assumed. Selecting a larger value for C will result in new observations having less influence over the Dirichlet distribution.

Additionally, it is important to define the collection of ratings for a source. The aggregate ratings for a source are stored as a cumulative vector expressed by (6). To update $\vec{R}_s$, the newly received rating vector $\vec{r}_s$ is added to the previously stored vector. It is possible to assign a higher weight to more recent ratings by incorporating a longevity factor $\lambda \in [0, 1]$, which controls the rate at which old

ratings are aged and degraded as a function of time. When $\lambda = 0$, ratings are forgotten after a single iteration, and ratings are never forgotten when $\lambda = 1$. This behavior is defined by (7), where $\vec{R}_{s,iter}$ represents the aggregate ratings of a source at iteration $iter$ and $\vec{r}_s$ represents a newly received source rating at iteration $iter + 1$.

$$\vec{R}_s = (R_s(i) | i = 1 \dots k) \quad (6)$$

$$\vec{R}_{s,iter+1} = \lambda \cdot \vec{R}_{s,iter} + \vec{r}_s, 0 \leq \lambda \leq 1 \quad (7)$$

The multinomial probability vector $\vec{S}$ for each k random probability variable is given by (8). When no ratings $\vec{r}_s$ have been made about a source, the aggregate rating vector $\vec{R}_s$ is initialized as $[0]^k$. In this case, since we assume uniform distributions, a source's initial multinomial probability vector will be $[1/k]^k$.

$$\vec{S} = (S_1, \dots, S_k) \quad \text{where} \quad S_i = \frac{R_s(i) + Ca(i)}{C + \sum_{j=1}^k R_s(j)} \quad (8)$$

The multinomial probability density function presents some requirements such as $\sum_{i=1}^k a(i) = 1$ and $\sum_{i=1}^k S(i) = 1$, where i represents the respective rating level, and $S(i)$ represents the multinomial probability score of i .

Despite the information the multinomial probability representation provides, substantial space can be required to display this data when k assumes larger values. A more compact representation of the reputation score uses a single value within a predefined interval (in our case, 0-1). This can be achieved by assigning a point value to each rating level i and then computing the normalized weighted point estimate score shown in (9).

Assuming k different rating levels with point values evenly distributed in the range $[0, 1]$, the point value for each level can be calculated by the expression $\frac{i-1}{k-1}$. Considering the previous example of a source with no ratings, (9) yields a point estimate of 0.5, which corresponds to the initial value for r_s in the system.

$$r_s = \sum_{i=1}^k \frac{i-1}{k-1} S_i \quad (9)$$

To track the behavior of sources over time (Design Goal 4) and predict future behavior, the reputation system is required to store data about each source. Three essential pieces of information must be maintained to achieve this goal: the aggregated ratings $\vec{R}_s$ of a source, its multinomial probability reputation score vector $\vec{S}$, and a history of received ratings $\vec{r}_s$ for each source. By maintaining this information, the reputation system can effectively monitor and evaluate the performance of sources over time, whilst allowing parts of the truth discovery process to be revisited when a source is found to be unreliable (Design Goal 5).

The information flow of the reputation module is presented in Algorithm 2. The input of the algorithm consists of the rating vector $\vec{r}_s$ with size k , along with the respective source identifier. The output of this module consists of an up-to-date reputation score r_s for the source.

TABLE 2. Parameter selection guidance.

Parameter	Name	Action	Impact
k	Rating Levels	Increase k	Slower reputation evolution. Reputation scores are more condensed and execution time increases.
		Decrease k	Faster reputation evolution. Reputation scores are less condensed and execution time decreases.
γ	Dampening Factor	Increase γ	Confidence scores become more dispersed along the $[0, 1]$ interval.
		Decrease γ	Confidence scores become more concentrated around 0.5.
ρ	Influence of Related Claims	Increase ρ	Higher influence of related claims when calculating the confidence score of a claim.
		Decrease ρ	Lower influence of related claims when calculating the confidence score of a claim.
λ	Longevity Factor	Increase λ	Higher weight given to older claims. If $\lambda = 1$, older claims are never forgotten.
		Decrease λ	Lower weight given to older ratings. If $\lambda = 0$, existing ratings are completely ignored.

Algorithm 2 Pseudocode for the Reputation Module

```

1: Input:  $\vec{r}_s, s$ 
2: Output:  $r_s$ 
3:
4: //Step 1: Check whether source  $s$  has previous contributions and fetch the respective reputation score
5: if  $\vec{R}_s$  exists then
6:    $\vec{R}_s \leftarrow$  fetch  $\vec{R}_s$  from the reputation module's database
7: else
8:    $\vec{R}_s \leftarrow$  initialize with  $[0]^k$ 
9: end if
10:
11: //Step 2: Add the new rating to the aggregate ratings  $\vec{R}_s$ 
12:  $\vec{R}_{s,iter+1} = \lambda \cdot \vec{R}_{s,iter} + \vec{r}_s$  /(Equation (7))
13:
14: //Step 3: Calculate the new multinomial probability vector
15:  $\vec{S} = (S_1, \dots, S_k)$  where  $S_i = \frac{R_s(i) + Ca(i)}{C + \sum_{j=1}^k R_s(j)}$  (Equation (8))
16:
17: //Step 4: Calculate the reputation score using a point estimate
18:  $r_s = \sum_{i=1}^k \frac{i-1}{k-1} S_i$  /(Equation (9))
19:
20: //Step 5: Update the database with the up-to-date  $r_s, \vec{S}$ , and  $\vec{R}_s$ 
21: db  $\leftarrow$  update database with  $\vec{R}_s, \vec{S}$ , and  $\vec{r}_s$ 
22:
23: return  $r_s$ 

```

D. PARAMETER CONFIGURATION GUIDELINES

Aiming to provide practitioners and researchers with guidance on parameter selection, Table 2 summarizes the impact of adjusting each of the key parameters of Nexus. By analyzing the described effects of each parameter, practitioners and researchers can identify the most appropriate configuration for their specific deployment scenario, establishing an informed starting point for parameter tuning that maximizes the accuracy of the data fusion process.

The number of rating levels k controls the rate at which source reputation evolves, with larger values of k increasing execution time for larger datasets. The dampening factor γ governs the dispersion of confidence scores within the interval $[0, 1]$. Higher values of γ increase dispersion, pushing confidence scores of claims provided by reputable sources with supportive relationships toward 1, and those provided by less reputable sources with contradictory relationships toward 0. Lower values of γ concentrate confidence scores around the initial value, reducing their spread. The parameter ρ controls the influence of relationships between related claims on confidence score computation. Higher values of ρ amplify the impact of both supportive and contradictory relationships, while lower values diminish it. Setting $\rho = 0$ removes the influence of inter-claim relationships entirely. The longevity factor λ determines the relative weight of historical ratings when updating source reputation scores. When $\lambda = 1$, all prior ratings are fully retained, whereas $\lambda = 0$ causes them to be entirely disregarded. Intermediate values of λ allow practitioners to balance the influence of historical and recent ratings according to the dynamics of their deployment scenario.

E. ADVERSARIAL ANALYSIS

Nexus introduces three parameters that influence the behavior of the data fusion process. The dampening factor γ is used to control overly high confidence scores by limiting the impact of source dependence. The number of levels k in the source rating vector determines the granularity of the reputation system, affecting how precisely source reputation can be represented and influencing the speed at which reputation scores evolve. Lastly, ρ controls the influence of related claims about the same individual when calculating the confidence score. Adjusting these parameters allows fine-tuning Nexus's performance to better suit specific application scenarios and counter potential adversarial behaviors. The selection of appropriate parameter values can significantly affect the accuracy and reliability of the data fusion process.

An expectable scenario in non-authoritative identity environments involves the presence of unreliable sources that may intentionally provide false information. Such adversarial

behavior can significantly affect the accuracy of the data fusion process if not properly addressed. In addition to the presence of a large number of unreliable sources, there is also the possibility of having low variability in the claims provided by sources. This situation can arise, for example, when sources copy information from one another or when adversaries coordinate a collusion attack, leading to a lack of claim diversity and, consequently, making it more challenging to identify the truth when a sufficiently large number of sources provide false or inaccurate claims.

Considering this high-noise, low-variability scenario, we present a mathematical analysis demonstrating how Nexus enables fine-tuning its parameters to mitigate the impact of such adversarial behaviors. In this context, the analysis focuses on the influence of ρ . We restrict our attention to categorical data, as it is the most susceptible to low variability. The analysis is conducted under the following assumptions: (i) there are only two possible claims about an individual, the true claim T and the false claim F ; (ii) all “good” sources providing T , denoted by G , have the same reputation r_G ; and (iii) all “bad” sources providing F , denoted by B , have the same reputation r_B . Under these assumptions, the goal is to determine ρ such that the confidence score of the true claim, $\phi(T)$, is always higher than that of the false claim, $\phi(F)$, even in scenarios where the number of bad sources (N_B) is significantly larger than the number of good sources (N_G) (high-noise scenarios). The assumptions ii and iii are subsequently relaxed to accommodate heterogeneous source reputations.

$$\begin{cases} \omega(T) = \sum_{i=1}^{N_G} -\ln(1 - r_G) = -N_G \cdot \ln(1 - r_G) \\ \omega(F) = \sum_{i=1}^{N_B} -\ln(1 - r_B) = -N_B \cdot \ln(1 - r_B) \end{cases} \quad (10)$$

The first step consists of calculating (10) and (11) for both claims T and F . Since we assume categorical data, the implication between claims in (11) is either 1 (when the claims are identical) or -1 (when they are conflicting).

$$\begin{cases} \sigma(T)[t] = \sum_{i=1}^{N_G} \omega(T) \cdot 1 + \sum_{i=1}^{N_B} \omega(F) \cdot -1 \\ \quad = N_G \cdot \omega(T) - N_B \cdot \omega(F) \\ \sigma(F)[t] = \sum_{i=1}^{N_B} \omega(F) \cdot 1 + \sum_{i=1}^{N_G} \omega(T) \cdot -1 \\ \quad = N_B \cdot \omega(F) - N_G \cdot \omega(T) \end{cases} \quad (11)$$

By substituting (10) and (11) into (3), we obtain the confidence scores for both claims. The objective of this analysis is to ensure that $\phi(T) > \phi(F)$, which yields the

inequality presented in (12).

$$\begin{aligned} \phi(T) &> \phi(F) \\ \Leftrightarrow \frac{1}{1 + e^{-\gamma \cdot (\omega(T) + \rho \cdot \sigma(T))}} &> \frac{1}{1 + e^{-\gamma \cdot (\omega(F) + \rho \cdot \sigma(F))}} \\ \Leftrightarrow \omega(T) + \rho \cdot \sigma(T) &> \omega(F) + \rho \cdot \sigma(F) \\ \Leftrightarrow \rho \cdot (\sigma(T) - \sigma(F)) &> \omega(F) - \omega(T) \end{aligned} \quad (12)$$

The final step consists of isolating ρ in the inequality presented in (12). This inequality results in three possible cases, depending on the value of $\sigma(T) - \sigma(F)$, as shown below.

$$\begin{cases} \sigma(T) - \sigma(F) > 0 \Rightarrow \rho > \frac{\omega(F) - \omega(T)}{\sigma(T) - \sigma(F)} \\ \sigma(T) - \sigma(F) < 0 \Rightarrow \rho < \frac{\omega(F) - \omega(T)}{\sigma(T) - \sigma(F)} \\ \sigma(T) - \sigma(F) = 0 \Rightarrow \omega(T) > \omega(F) \end{cases}$$

The first case, where $\sigma(T) - \sigma(F) > 0$, depicts an ideal scenario, where the influence of true claims outweighs that of false claims, thus not being the focus of this analysis. In the high-noise scenario under consideration, the number of bad sources (N_B) exceeds the number of good sources (N_G). As a result, it is expected that $\sigma(F) > \sigma(T)$, which implies that $\sigma(T) - \sigma(F) < 0$. Consequently, to guarantee that $\phi(T) > \phi(F)$, the second case must be satisfied. The inequality that ρ must fulfill in this situation is given in (13). Lastly, the third case, where $\sigma(T) - \sigma(F) = 0$, is not the focus of this analysis since it solely relies on the reputation of sources providing each claim (as demonstrated in (10)), with ρ having no influence on the outcome.

$$\begin{aligned} \rho &< \frac{\omega(F) - \omega(T)}{\sigma(T) - \sigma(F)} \\ \Leftrightarrow \rho &< \frac{N_G \ln(1 - r_G) - N_B \ln(1 - r_B)}{2(N_G \omega(T) - N_B \omega(F))} \\ \Leftrightarrow \rho &< \frac{N_G \ln(1 - r_G) - N_B \ln(1 - r_B)}{2(N_G^2 \ln(1 - r_G) - N_B^2 \ln(1 - r_B))}, \end{aligned} \quad r_G, r_B \in]0, 1[\quad (13)$$

Equation (4) establishes a mathematical foundation for choosing an appropriate value of ρ in scenarios with high noise and low variability. By satisfying this inequality, Nexus can effectively limit the impact of unreliable sources, ensuring that the confidence score of the true claim remains higher than that of the false claim ($\phi(T) > \phi(F)$), even when the number of bad sources exceeds the number of good sources ($N_B > N_G$).

Figure 4 depicts the evolution of the upper bound for ρ as a function of r_G and r_B across four different noise scenarios. Each subfigure illustrates a distinct combination of N_G and N_B , highlighting how the reputations of good and bad sources affect the selection of ρ in high-noise environments. It also enables the assessment of the maximum value of ρ that

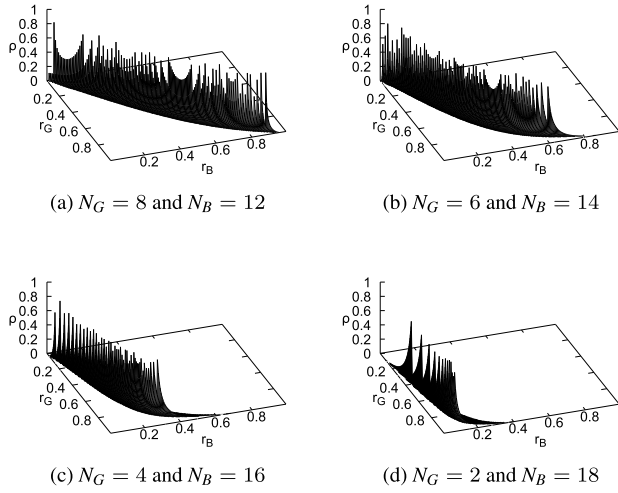


FIGURE 4. Evolution of the upper bound for ρ as a function of r_G and r_B in different noise scenarios.

maintains the condition $\phi(T) > \phi(F)$, thereby preserving the integrity of the data fusion process. For instance, a key takeaway of this analysis resides in an increase in the number of bad sources (N_B), resulting in a lower maximum allowable reputation (r_B) for each bad source.

A natural extension of the preceding analysis consists of relaxing assumptions (ii) and (iii) by allowing source reputations to vary individually, rather than enforcing a uniform reputation across all sources supporting the same claim. This generalization yields a more realistic model, as sources corroborating the true claim T may exhibit heterogeneous reliability levels, and the same applies to sources supporting the false claim F . Under this generalized formulation, the claim weights ω are no longer given by $-N_G \ln(1 - r_G)$ and $-N_B \ln(1 - r_B)$ instead, they are expressed as shown below, where $r_{G,i}$ and $r_{B,j}$ denote the individual reputations of the i -th good source and the j -th bad source, respectively.

$$\omega(T) = \sum_{i=1}^{N_G} -\ln(1 - r_{G,i}) \quad \omega(F) = \sum_{j=1}^{N_B} -\ln(1 - r_{B,j})$$

Although source reputations could be summarized using their arithmetic mean, this is not the most appropriate approach, since the contribution of each source to ω is not proportional to its reputation. In other words, the model does not combine reputations linearly, so replacing all individual reputations by their arithmetic mean may distort the final result. For this reason, we use an effective reputation denoted by $\tilde{r}_G$ for good sources (and $\tilde{r}_B$ for bad sources) which is defined to produce exactly the same total proof weight as the original set of heterogeneous source reputations. Drawing from (10), we derive the effective reputation $\tilde{r}_G$ for good sources in (14), noting that the same process can be applied

to derive $\tilde{r}_B$ for bad sources.

$$\begin{aligned} -N_G \cdot \ln(1 - \tilde{r}_G) &= \sum_{i=1}^{N_G} -\ln(1 - r_{G,i}) \\ \Leftrightarrow N_G \cdot \ln(1 - \tilde{r}_G) &= \sum_{i=1}^{N_G} \ln(1 - r_{G,i}) \\ \Leftrightarrow \ln(1 - \tilde{r}_G) &= \frac{1}{N_G} \cdot \sum_{i=1}^{N_G} \ln(1 - r_{G,i}) \\ \Leftrightarrow 1 - \tilde{r}_G &= e^{\frac{1}{N_G} \cdot \sum_{i=1}^{N_G} \ln(1 - r_{G,i})} \\ \Leftrightarrow \tilde{r}_G &= 1 - e^{\frac{1}{N_G} \cdot \sum_{i=1}^{N_G} \ln(1 - r_{G,i})} \end{aligned}$$

$$r_{G,i} \in]0, 1[\quad (14)$$

Having derived both $\tilde{r}_G$ and $\tilde{r}_B$, (13) can be updated by replacing r_G and r_B with their effective counterparts, resulting in (15), which yields the same upper bound for ρ as in the original analysis. This extension allows for a more realistic assessment of the impact of ρ in scenarios where source reputations are heterogeneous, providing a more comprehensive understanding of how to fine-tune Nexus's parameters to mitigate adversarial behaviors effectively.

$$\rho < \frac{N_G \ln(1 - \tilde{r}_G) - N_B \ln(1 - \tilde{r}_B)}{2(N_G^2 \ln(1 - \tilde{r}_G) - N_B^2 \ln(1 - \tilde{r}_B))}, \quad \tilde{r}_G, \tilde{r}_B \in]0, 1[\quad (15)$$

This analysis provides evidence that Nexus is designed to mitigate the impact of adversaries by enabling fine-tuning of its parameters. A further exploration of adversarial scenarios and their implications for Nexus's behavior is left for future work. An assessment of this nature is crucial for understanding Nexus's resilience against various types of attacks and, therefore, for ensuring its adequacy in the context of non-authoritative identity systems.

IV. PROOF OF CONCEPT

A publicly available implementation of Nexus² was assessed to demonstrate its accuracy and capability of consolidating divergent claims (Design Goal 2) in the context of non-authoritative identity. The evaluation is organized into two scenarios: *Scenario 1* examines Nexus in a controlled environment using publicly available synthetic and curated data to assess its fundamental behavior and performance against two established truth discovery algorithms, while *Scenario 2* demonstrates Nexus's applicability by testing the public implementation using real-world records from a low-income country health service. Additionally, *Scenario 1* includes an analysis of how the parameters γ , ρ , and k influence confidence scores and reputation evolution.

²Implementation available at <https://github.com/dissect-project/nexus>

All experiments were performed on a Dell OptiPlex 5000 with an Intel i5-12500 processor, with 256GB NVMe storage and 64GB memory, running Rocky Linux 9.4. All baseline methods, as well as Nexus, are implemented using Python 3.12.3. The results for both scenarios are presented and discussed in Section V.

Accuracy is evaluated using several distance and similarity functions by computing the normalized similarity between the output of each method and the ground truth (also known as the gold standard), defined as the set of verified facts against which the system's performance is benchmarked. It is important to note that the gold standard is used solely to assess the solution's performance and is neither processed nor utilized during the truth discovery process. Jaro-Winkler, Sørensen-Dice, and Damerau-Levenshtein are used for string data [52], [56], [57], [58], whilst Euclidean and Manhattan distances are used for continuous data [32]. For categorical values, a direct comparison method is used, where matching values return a score of 1 and contradicting values return a score of -1 . The reported execution time and accuracy values represent the averages calculated from 10 independent evaluation runs. To avoid the possibility of the order of execution influencing the results, the processing of values by Nexus is randomized before each execution. The testbed used in the experiments is available at a public repository³ with all the information required to ensure full reproducibility of the results presented in Section V.

A. SCENARIO 1: CONTROLLED ENVIRONMENTS

In this scenario, we evaluate Nexus using a suite of synthetic and curated datasets designed to simulate conditions typically found in identity systems. This controlled setting allows us to isolate variables and rigorously assess Nexus's performance, stability, and parameter sensitivity under varying levels of noise and data quality. Additionally, the performance of Nexus is benchmarked against two established truth discovery algorithms, TruthFinder and CRH. These algorithms were selected as the reference baselines since, among those presented in Section II, they are the only ones capable of handling heterogeneous data, making them the most appropriate candidates for a meaningful comparison in the context of identity systems.

Table 3 provides an overview of the data employed in this evaluation,⁴ which comprises both synthetic and curated datasets. The synthetic datasets were generated to systematically assess Nexus's performance across individual data types, emulating common error patterns found in digital identity systems, such as typos or factual inaccuracies. Throughout this evaluation, these deviations and inconsistencies are collectively referred to as *noise*. In contrast, the curated datasets were selected to represent practical scenarios where Nexus operates within non-authoritative

identity environments, utilizing public data commonly used in the digital identity research field.

1) SYNTHETIC DATASETS

To encompass all data types identified in Section II, we constructed four synthetic datasets: *synth-cat*, *synth-str*, *synth-con*, and *synth-het*. The *synth-cat* dataset includes a variety of nationalities, where noise corresponds to any country differing from the true value. The *synth-str* dataset comprises full names of individuals, with noise introduced through typographical errors, spelling variations, and omissions of one or more names. The *synth-con* dataset consists of numerical variables representing ages, where noise is incorporated through random deviations from the true value. Finally, the *synth-het* dataset integrates the data types and respective noise characteristics of all three aforementioned datasets, thereby representing mixed-data scenarios that reproduce more closely real-world conditions.

For each dataset, multiple scenarios were generated by varying the number of sources and the degree of noise introduced. Specifically, each dataset was instantiated with 5, 10, 15, and 20 sources. For each source scenario, four noise levels were considered: 20%, 40%, 60%, and 80%. For instance, a scenario with 10 sources and 40% noise includes six reliable sources (60%) and four noisy sources (40%), *i.e.*, sources providing claims that differ from the true value. Conversely, reliable sources are those that consistently provide accurate information matching the ground truth.

The only exception to this setup is the scenario with 5 sources, where the 80% noise scenario was omitted, as having a case with only one “accurate” source (20%) would not yield a meaningful evaluation. In total, this resulted in 15 distinct scenarios per dataset type. The number of claims in each scenario is equal to the number of sources, as each source provides a single claim for each object. The exception to this rule is the *synth-het* dataset, where each source provides claims for all three data types, resulting in a total of three claims per source.

For both synthetic and curated datasets, the parameters γ and ρ were set to 0.15 and 0.9, aiming to provide a balance between claim confidence dispersion and evolution. The longevity factor λ was set to 1 for both datasets as we do not want previous contributions to be “forgotten” over time. The number of levels k in the source rating vector was set to 10 for the synthetic datasets, as the number of claims per source is relatively low compared to the curated datasets. This configuration ensures a granular representation of source reputation while preventing overly rapid evolution of scores.

2) CURATED DATASETS

To evaluate Nexus in scenarios that more closely resemble real-world applications for non-authoritative identity, three publicly available datasets were selected: *Adult* [59], *Diabetes* [60], and *Gender* [61]. These datasets were retrieved

³Testbed available at <https://github.com/dissect-project/nexus-testbed>

⁴Datasets available at <https://github.com/dissect-project/nexus-datasets>

TABLE 3. Summary of datasets and configurations used in Scenario 1.

	Dataset	Data Types	Claims	Sources	Noise Levels	Configuration
Synthetic	synth-cat	Categorical	5	5	20%, 40%, 60%	$k = 10$ $\gamma = 0.15$ $\rho = 0.9$ $\lambda = 1.0$
			10, 15, 20	10, 15, 20	20%, 40%, 60%, 80%	
	synth-str	String	5	5	20%, 40%, 60%	
			10, 15, 20	10, 15, 20	20%, 40%, 60%, 80%	
	synth-con	Continuous	5	5	20%, 40%, 60%	
			10, 15, 20	10, 15, 20	20%, 40%, 60%, 80%	
	synth-het	Categorical, String, Continuous	15	5	20%, 40%, 60%	
			30, 45, 60	10, 15, 20	20%, 40%, 60%, 80%	
Curated	Adult	Categorical, String, Continuous	3,907,320	20	20%, 40%, 60%, 80%	$k = 20$
	Diabetes	Categorical, String, Continuous	8,483,040	20	20%, 40%, 60%, 80%	$\gamma = 0.15$ $\rho = 0.9$
	Gender	Categorical, String, Continuous	8,835,840	20	20%, 40%, 60%, 80%	$\lambda = 1.0$

from the UC Irvine Machine Learning Repository⁵ and were chosen due to their inclusion of heterogeneous data types, encompassing categorical, string, and continuous variables. Each dataset shares the same number of sources (i.e., 20) and noise levels (20%, 40%, 60%, and 80%).

The *Adult* dataset comprises demographic information for 32,561 unique individuals. From the available attributes, six were selected: age (continuous), work class (categorical), education (categorical), marital status (categorical), country (string), and working hours per week (continuous). There are 20 sources providing claims for each individual, resulting in a total of 3,907,320 claims. A claim consists of source-provided values for an individual's attribute. Regarding the introduced noise, categorical values were altered by randomly selecting a different value, string values were modified by replacing between 1 and $\frac{\text{length}}{5}$ characters with different characters, and continuous values were adjusted by adding or subtracting a random non-zero value between -20 and 20.

The *Diabetes* dataset contains medical records for 70,692 unique individuals. From the available attributes, six were selected: diabetes (categorical), high blood pressure (categorical), high cholesterol (categorical), age bracket (categorical), body mass index (continuous), and education (string). There are 20 sources providing claims for each individual, resulting in a total of 8,483,040 claims. The noise generation process is similar to that of the previous dataset for categorical values. For continuous values, noise was introduced by selecting a random value outside the range of possible values for that attribute (the range of values for body mass index is contained between 10 and 60). Finally, string values were modified by randomly replacing between 2 and $\frac{\text{length}}{5}$ characters.

⁵<https://archive.ics.uci.edu>

Lastly, the *Gender* dataset combines raw counts for first or given names of babies born in the United States, United Kingdom, Canada, and Australia, spanning multiple decades. The select attributes for this dataset include the name (string), gender (categorical), and the aggregate count of births for the name (continuous). In total, there are 20 sources providing claims for 147,264 unique name and gender combinations, resulting in a total of 8,835,840 claims. The noise generation process follows the same methodology as the previous datasets for categorical and string values. For continuous values, noise was introduced by adding or subtracting a random integer in the range $[0, \text{value}]$, where *value* represents the continuous claim itself.

For these datasets, the number of levels k in the source rating vector was set to 20, as the number of claims per source is relatively high compared to the synthetic datasets. This configuration ensures a granular representation of source reputation while preventing overly rapid evolution of scores.

B. SCENARIO 2: REAL-WORLD APPLICABILITY

In this scenario, we assessed the applicability and effectiveness of Nexus in a real-world setting where data quality and source reliability vary significantly. Specifically, we conducted an experiment using data collected over one year of calls to Guinea-Bissau's 24-hour Health Line service, operated by the Non-Governmental Organization (NGO) VIDA.⁶ This context is particularly relevant given that the overall birth registration rate in Guinea-Bissau currently stands below 50%, with approximately 46% of children under the age of five registered, implying that over half of

⁶The project "24h Health Line - Strengthening the availability and proximity of Health Services to the population of Guinea-Bissau" is implemented by the NGDO VIDA in partnership with the Ministry of Public Health of Guinea-Bissau, with funding from Camões IP, UNDP Guinea-Bissau, UNICEF, and the United Nations Peacebuilding Fund.

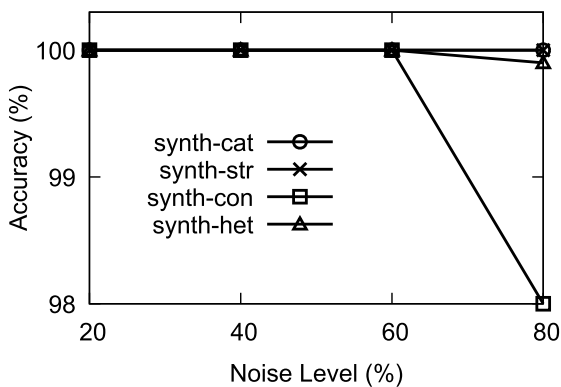
TABLE 4. Summary of the real-world dataset.

Data Types	Claims	Sources	Configuration
Categorical, Continuous, String	57691	14	$k = 5$ $\gamma = 0.5$ $\rho = 0.9$ $\lambda = 1.0$

this demographic lacks a legal identity registry, placing the country among those most critically affected by the lack of legal identity coverage [62]. The dataset comprises records of individuals who contacted the health line service, providing various personal details across multiple interactions. As summarized in Table 4, it contains 5,444 unique individuals and claims collected from 14 sources, each corresponding to a distinct health line operator responsible for recording patient information during telephone interactions.

A patient record in this dataset may contain at most 12 attributes, including categorical data (e.g., gender), continuous data (e.g., age), and string data (e.g., name or address). However, not all records contain all attributes, as some patients may not provide information about certain attributes, or operators may omit or incorrectly record some details.

As patients may contact the health line on multiple occasions, several records may exist for the same individual, each potentially containing varying or conflicting information. This scenario is particularly relevant in LICs, where individuals may lack official identification documents, leading to previously explored challenges related to accurately consolidating identity information.


FIGURE 5. Synthetic datasets.

Considering the smaller dimensions of this dataset compared to the curated datasets used in the previous sections, Nexus was configured with the settings described in Table 4. These parameters allow sources to have a faster reputation evolution, which is suitable for this case, as each source has, on average, fewer than 4000 claims. The dampening factor γ was set to 0.5 to provide a balanced approach between claim confidence dispersion and evolution. The influence of related claims ρ was set to 0.9 to take advantage of

existing relationships between claims, given the assumption that reliable sources will support each other. As with previous tests, claims were randomly shuffled before being consolidated.

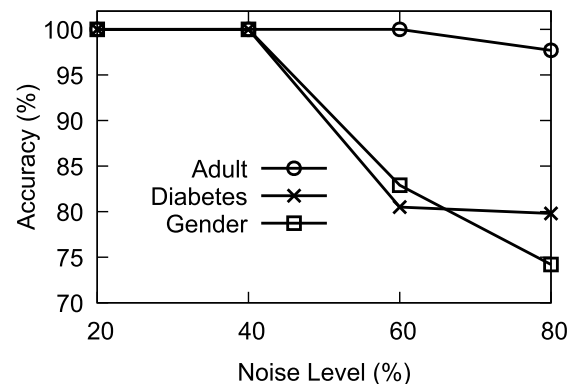
V. RESULTS

This section presents the experimental results from evaluating Nexus across the two described scenarios. First, we analyze the system's performance in a controlled environment using synthetic and curated datasets (Scenario 1), focusing on its sensitivity to noise and parameter configurations, and measuring its performance against TruthFinder and CRH. Second, we assess its real-world applicability using a real digital identity dataset (Scenario 2) to demonstrate its effectiveness in practical settings.

A. SCENARIO 1: CONTROLLED ENVIRONMENTS

Table 5 summarizes the average similarity results for the synthetic datasets across all configurations (see Table 3). The results indicate that Nexus achieves high accuracy levels across all data types, with average similarity scores of 1.0 for categorical and string data, 0.996 for continuous data, and 0.999 for heterogeneous data. These findings demonstrate the effectiveness of Nexus in accurately consolidating claims from multiple sources, even in the presence of significant number of claims that are either conflicting or contain noise.

A more detailed analysis of these results is presented in Figure 5, which illustrates the evolution of accuracy as noise levels increase for each synthetic dataset. The results show that Nexus maintains high accuracy levels even as noise increases, with accuracy only suffering a small decrease in the presence of 80% of sources providing noisy or incorrect claims. Moreover, the decrease in accuracy is more pronounced in the *synth-con* and *synth-het* datasets, which can be attributed to the nature of continuous data and the methodology used to perform the experiments.


FIGURE 6. Curated datasets.

The continuous data present in these datasets consists of age values, which have a limited range of possible values (from 25 to 35). Additionally, the noise generation process for continuous data involves adding or subtracting random

TABLE 5. Average similarity for the synthetic datasets.

Dataset	Algorithm						Average Similarity
	Direct Comparison	Jaro-Winkler	Sorensen-Dice	Damerau-Levenshtein	Euclidean	Manhattan	
synth-cat	1.0	-	-	-	-	-	1.0
synth-str	-	1.0	1.0	1.0	-	-	1.0
synth-con	-	-	-	-	0.996	0.996	0.996
synth-het	1.0	1.0	1.0	1.0	0.999	0.999	0.999

values within the range of $[-5, 5] \setminus \{0\}$. This results in noisy claims that are often close to the true value, making it more challenging for Nexus to accurately distinguish between true and noisy claims, as the resulting $\phi(c)$ values become more condensed. This concentration in the claims provided by sources was specifically designed to mimic real-world scenarios, where noisy data is not always drastically different from the truth, that is, different sources may provide similar claims about an individual having minor deviations from the ground truth.

Considering the *synth-con* dataset (as it was the worst performing one) configurations with 80% noise (10, 15, and 20 sources), and considering that each dataset was executed 5 times, in total, there were 15 executions. From these 15 executions, 6 resulted in exact matches, and 9 resulted in an average (and maximum) error of 1 year when compared to the true value. This indicates that even in scenarios where Nexus is not capable of identifying the correct value, the deviation from the truth remains relatively low.

Despite presenting good results for synthetic datasets, it is important to evaluate Nexus in real-world scenarios to validate its effectiveness in practical applications, as good results with synthetic workloads do not guarantee the same real-world performance. Table 6 summarizes the average similarity results for the curated datasets across all noise levels, whilst Figure 6 presents the evolution of Nexus' average accuracy across all noise levels.

Results show that Nexus is capable of achieving high accuracy levels for low and medium noise levels (*i.e.*, 20% and 40%) across all datasets. However, as noise levels increase to 60% and 80%, a more significant drop in accuracy is observed, particularly in the *Diabetes* and *Gender* datasets. This decline can be attributed to the decline in accuracy for categorical data. In the case of the *Diabetes* dataset, there are 4 categorical attributes, and from these, 3 present only binary values (*i.e.*, yes/no). Moreover, the *Gender* dataset contains a single categorical attribute which also presents two possible values ("M" and "F"). This reinforces the previous finding that scenarios with a high percentage of sources providing noisy claims, combined with a reduced set of variability in data, are the main reason for decreases in accuracy for Nexus.

Regarding this particular aspect, it is important to note that Nexus' configuration allows for fine-tuning the influence of related facts through parameter ρ . By adjusting this

parameter, users can modulate the weight assigned to related claims when calculating the adjusted confidence score ϕ (as defined in (3)). Considering the case of the *Gender* dataset with 80% noise, and considering a case where the true value is "F", there will be 4 sources providing the accurate claim "F" and 16 sources providing the noisy claim "M". By setting ρ to a higher value (closer to 1), the influence of related claims is amplified. Even if the 4 sources providing the true value have a higher reputation (and consequently, claims provided by them will have a higher weight ω), the remaining 16 sources will outweigh them (due to (3)), as all of the 16 claims support each other. This behavior was mathematically analyzed in Section III-E, specifically, in Figure 4c, where it is possible to observe that the selected value of ρ (0.9) is above the maximum allowable value to ensure that $\phi(T) > \phi(F)$ in this scenario. Conversely, lowering ρ (closer to 0) reduces the influence of related claims, which can help mitigate the impact of a large number of noisy sources providing the same incorrect claim.

To demonstrate this behavior, we reassessed the *Diabetes* dataset with ρ set to 0.1 (instead of 0.9). The results are presented in Table 7 and show that for higher noise levels (60% and 80%), lowering the influence of related claims does improve the accuracy for categorical data. However, in doing so, string and continuous data show a decrease in accuracy. This can be attributed to the fact that with these two data types, there is more variation, and hence, the supportive relation between claims plays a significant role in the truth discovery process. With this analysis, it becomes clear that different parameter configurations can be employed to adapt Nexus to the scenario in which it is deployed. The parameter sensitivity analysis is explored later in this section, including an evaluation on ρ , k and γ .

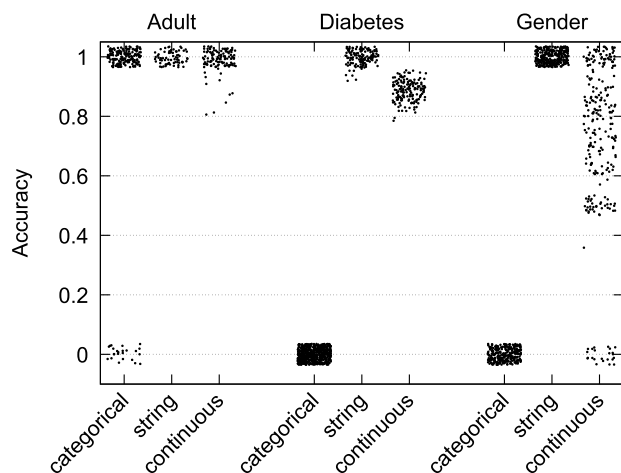
Even though Nexus suffers a decrease in accuracy for categorical data in low-variability scenarios with high noise levels, it is important to highlight that the overall average similarity remains high across all tested datasets. From the analysis present in Figure 6, it is possible to observe that with 60% noise, the average accuracy across all datasets remains above 80%, and even with 80% noise, the lowest average accuracy is approximately 74%. Figure 7 presents a more detailed view of the results in terms of accuracy for all curated datasets regarding the scenario with 80% noise, illustrating the distribution of accuracy scores for all consolidated claims

TABLE 6. Average similarity for the curated datasets by noise level.

Dataset	Noise	Algorithm						Average Similarity
		Direct Comparison	Jaro-Winkler	Sorensen-Dice	Damerau-Levenshtein	Euclidean	Manhattan	
Adult	20%	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	40%	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	60%	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	80%	0.881	1.0	1.0	1.0	0.987	0.987	0.98
Diabetes	20%	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	40%	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	60%	0.21	0.999	0.999	0.999	0.951	0.951	0.85
	80%	0.0	0.994	0.995	0.995	0.887	0.887	0.79
Gender	20%	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	40%	1.0	1.0	1.0	1.0	1.0	1.0	1.0
	60%	0.0	1.0	1.0	1.0	0.942	0.942	0.81
	80%	0.0	0.999	0.999	0.999	0.718	0.718	0.74

TABLE 7. Result for the *Diabetes* dataset with $\rho = 0.1$.

Algorithm	Noise Level			
	20%	40%	60%	80%
Direct Comparison	0.80	0.60	0.40	0.20
Jaro-Winkler	0.97	0.95	0.92	0.89
Sorensen-Dice	0.97	0.95	0.92	0.89
Damerau-Levenshtein	0.96	0.93	0.89	0.85
Euclidean	0.97	0.95	0.92	0.89
Manhattan	0.97	0.95	0.92	0.89

**FIGURE 7.** Accuracy for curated datasets with 80% noise.

across all executions. Given the high number of claims (over 8 million for the *Gender* dataset), random sampling of 2% was employed to produce a readable but representative plot.

Considering the *Adult* dataset, results show that the accuracy of most claims is concentrated around 1.0 whilst exhibiting low variability, with only a few claims exhibiting

lower accuracy values (mostly categorical data). With the *Diabetes* dataset, a similar pattern of low variability is observed, with string and continuous claims showing accuracy values close to 1.0. The exception to this resides in categorical claims, which are concentrated around 0.0 due to combination of high noise levels and low variability in the data (as previously discussed). In the case of the *Gender* dataset, a similar trend can be observed for categorical claims. On the other hand, string claims exhibit a high concentration of accuracy values close to 1.0, whilst having a low degree of variability. Finally, continuous claims show a wider distribution of accuracy values, mostly concentrated in the [0.5, 1.0] interval, with a few outliers exhibiting lower accuracy values. The wider distribution of accuracy values for continuous claims can be attributed to the nature of the data and the noise generation process. The continuous value present in the *Gender* dataset corresponds to the aggregate count of births for a given name, which can be a very large number (in the order of millions). The noise generation process involves adding or subtracting a random value in the range $[0, \text{value}]$, which can lead to significant deviations from the true value (e.g., 10,000,000 and 100,000), thus resulting in a broader range of accuracy scores.

The results presented in Figures 6 and 7 allow us to draw several important conclusions regarding the performance of Nexus in high-noise scenarios, which is of particular interest when considering non-authoritative environments. Firstly, Nexus demonstrates a strong capability to accurately consolidate claims, as average scores are above 0.74 for all datasets in the worst-case scenario of 80% noise. Second, the cases where Nexus struggles to provide accurate results are primarily associated with categorical data in scenarios characterized by both high noise levels and low variability in the data. As ρ is set to 0.9, the inequality presented in (13) does not hold. This insight is crucial for understanding the importance of selecting adequate values for Nexus' parameters (ρ , k , and γ) and highlights areas

for further research regarding parameter fine-tuning. Third, the opposite scenario, where high variability in the data exists alongside high noise levels, increases the spread in accuracy scores, particularly for continuous data. This reinforces the finding that, while Nexus presents itself as an effective solution, data characteristics significantly influence its performance, underscoring the importance of appropriate parameter selection. Lastly, in the remaining cases, Nexus consistently maintains high accuracy with low variability, highlighting its robustness and reliability across diverse data types and noise conditions.

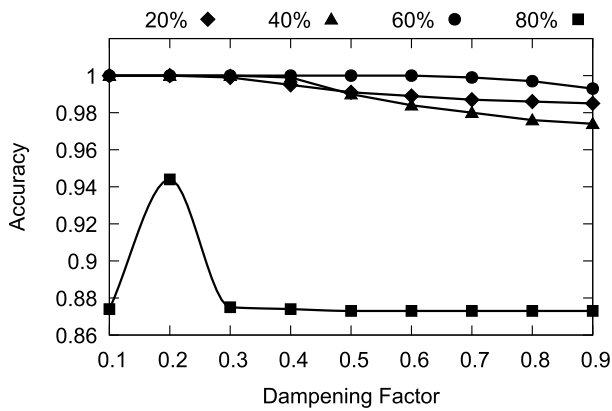


FIGURE 8. Accuracy for different values of γ .

A key aspect of Nexus is its customizability, which allows fine-tuning of parameters for both the truth discovery and reputation modules. As previously demonstrated, adjusting these parameters can significantly impact the system's performance. To demonstrate this, we performed a series of tests using a subset of *Adult* dataset, presenting the results as an average between five executions for each noise level. The subset used contains 5,000 unique individuals of the dataset, resulting in a total of 600,000 claims. Moreover, the same noise levels as in the previous experiments were used (20%, 40%, 60%, and 80%).

The first assessed parameter is the dampening factor γ . For this test, ρ was set to 1.0, and k was set to 20. Multiple tests were performed with increments to γ of 0.1, ranging from 0.1 to 0.9. The results shown in Figure 8 highlight that in this case, changing the dampening factor impacts the overall accuracy of Nexus. Having the dampening factor set to lower values (e.g., approximately 0.2) yields the highest average accuracy for all noise levels, whilst increasing γ to values in the range [0.3, 0.9] results in a gradual decrease in the overall accuracy. This behavior can be attributed to the way the confidence score ϕ is calculated (presented in (3)) and how different γ values influence the weight of $\omega(c) + \rho \cdot \sigma(c)$ (as demonstrated in Figure 9).

The main impact of lowering the dampening factor resides in the resulting confidence ϕ being closer to 0.5. Inaccurate values end up having a higher confidence score (e.g., 0.4 instead of 0.2), and accurate values have a lower

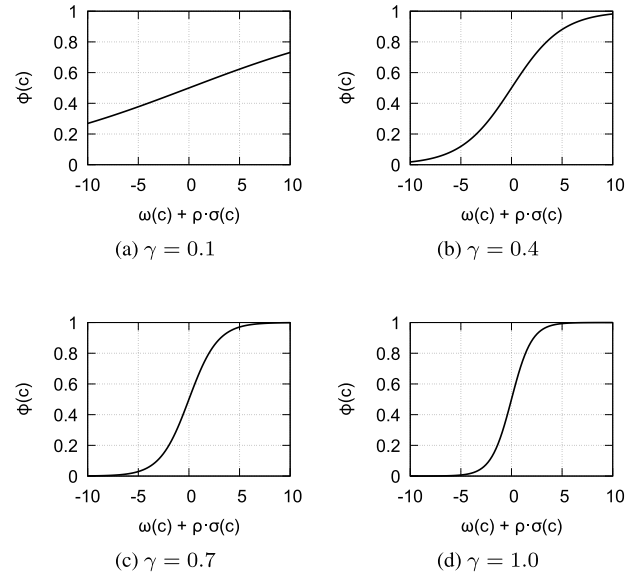


FIGURE 9. ϕ variation for different γ values.

confidence score (e.g., 0.6 instead of 0.8). In other words, by lowering the dampening factor, the interval of possible values for the confidence score is shortened (i.e., the results become more condensed around 0.5), as demonstrated in Figure 9a. This is a result of applying a sigmoid function [53] in (3). On the other hand, increasing the dampening factor results in higher confidence scores of accurate values and, conversely, lower scores of inaccurate values, as seen in Figure 9d. With a higher damping factor, the resulting confidence scores become more dispersed along the [0, 1] interval. The plots in Figure 9 depict the behavior of (3) for different γ values. The primary reason for reducing the dampening factor is to regulate the rate at which the scores of values with multiple supporting sources increase, thus avoiding having multiple claims all with maximum confidence scores.

This behavior is the main reason why the accuracy decreases when γ is set to 0.3 or higher in Figure 8. With higher dampening factors, multiple claims end up having maximum confidence scores. If several claims hit the ceiling of the confidence score range, the truth discovery algorithm struggles to differentiate between them, as all claims have the same confidence score, leading to the decrease in accuracy seen in this analysis.

The second evaluated parameter is ρ , which defines the weight of $\sigma(c)$. Tweaking this parameter allows us to define how related claims are taken into account when calculating the confidence score ϕ (as defined in (3)). The tests for this parameter were executed in the interval [0, 1], with increments to ρ of 0.1. Parameters k and γ were set to 20 and 0.15, respectively.

As shown in Figure 10, setting ρ in the interval [0.3, 0.5] presents the highest overall accuracy for the 80% noise, whilst the remaining levels remain stable, as they maintain 100%

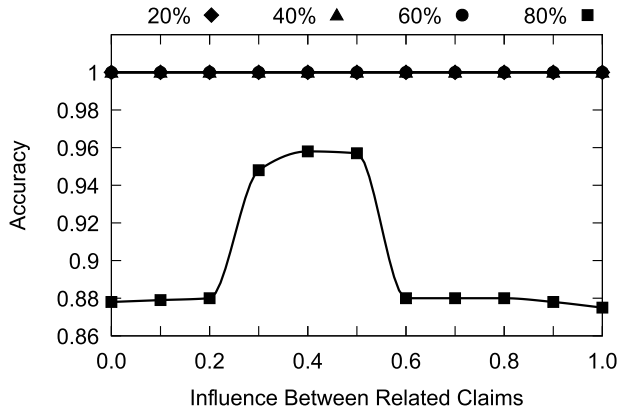


FIGURE 10. Accuracy for different values of ρ .

accuracy within all tested values. The explanation for the increase in accuracy for the 80% noise level resides in the categorical data, as that specific type of data was the only significantly affected. Increasing ρ to a high value (e.g., above 0.5) allows taking advantage of existing relationships between values (supportive or contradictive relationships). However, considering the analysis presented in Section III-E, with such a high noise level, ρ must be kept below the threshold defined in (13) to ensure that accurate values maintain a higher confidence score than inaccurate ones. In this case, the analysis of Figure 10 suggests that the threshold for ρ in this scenario is approximately 0.5.

The final parameter tested was k , which differs from the previous parameters as it is part of the reputation system rather than the truth discovery algorithm. As explained in Section III, k defines how many rating levels there are in the reputation system for each source rating $\vec{r}$. The tests for this parameter have been conducted in the interval [10, 250] with increments of 20. γ and ρ were set to 0.2 and 0.4, respectively.

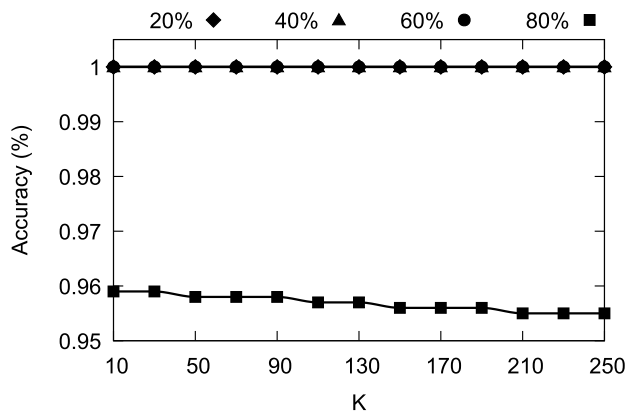


FIGURE 11. Accuracy for different levels of k .

Regarding the accuracy of the outputted values, Figure 11 shows that increasing the value of k does not provide any significant change. Throughout the entire range that has been tested, the accuracy of the results shows signs of stability,

with minor oscillations in the 80% noise level, where a small decrease is observed. Since k directly influences the size of $\vec{S}$, this will directly impact the time the reputation systems require to calculate and update the reputation scores of the sources. Aiming to evaluate this impact, the execution time was measured and presented in Figure 12. Results show that increasing k leads to a linear increase in the execution time.

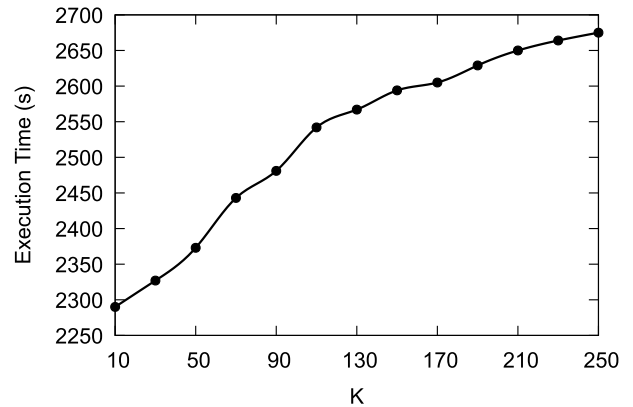


FIGURE 12. Execution time for different levels for k .

Alongside the increased execution time, there is a visible impact that is not directly related to any performance measure. Changing the value of k impacts the resulting reputation scores of sources since it dictates the number of ratings required for scores to increase or decrease, i.e., the speed at which reputation scores evolve. This behavior is expected due to the usage of a Dirichlet distribution [63] in the reputation system.

A lower value for k results in more dispersed reputation scores in the [0, 1] interval. Sources with high reputation scores are closer to 1, and conversely, sources with a low reputation score are closer to 0. However, if k is set to a higher value (e.g., 250), the reputation scores of sources will be more condensed around 0.5 (the initial reputation score for a source). This behavior is useful to control the rate at which source reputation evolves over time. Figure 13 showcases the evolution of a source's reputation score for k set to 10 and 250 in two distinct scenarios: a scenario where all received ratings are the best possible (Figure 13a), and a scenario where all received ratings are the worst possible (Figure 13b). This analysis allows understanding the evolution in the cases where a source is continuously evaluated with the highest and worst ratings possible, concluding that k can be used to tweak the rate of reputation evolution in the system, albeit at some cost in performance (as seen in Figure 12).

An analysis of continuous best and worst ratings provides a useful insight into the theoretical upper and lower bounds of reputation scores which a source can achieve for different k values. However, it is also important to understand how the reputation scores vary in a more realistic scenario. To demonstrate this, we selected three sources with the

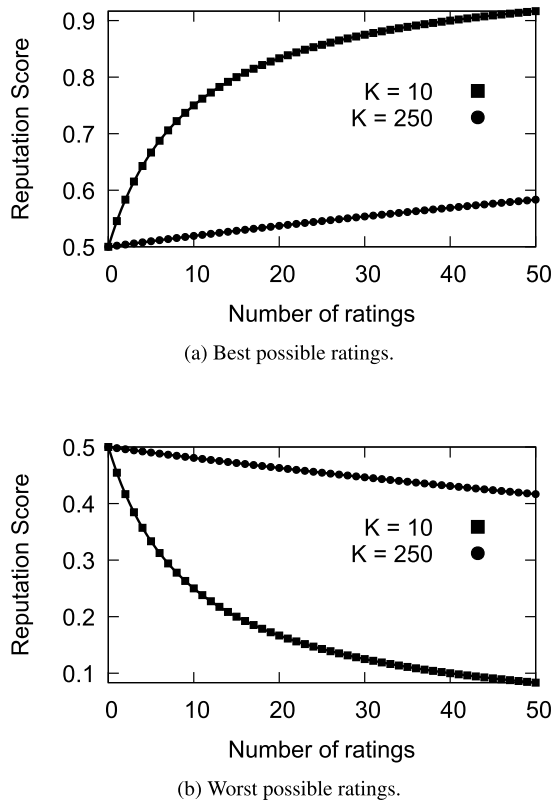


FIGURE 13. Reputation score variation for different k values.

TABLE 8. Reputation scores of most reliable sources.

Noise Level	Source	$k = 10$	$k = 250$
20%	source 17	0.9998	0.9955
	source 9	0.9998	0.9955
	source 19	0.9998	0.9955
80%	source 18	0.6344	0.6232
	source 19	0.6344	0.6232
	source 20	0.6344	0.6232

highest reputation scores for the 20% and 80% noise levels, presenting their scores for k set to 10 and 250 in Table 8.

Results for the 20% noise level demonstrate that with a lower k value (in this case, 10), the reputation scores of the most reliable (or best) sources amount to an average 0.9998, whereas with a higher k value (250), on average, the reputation scores amount to 0.9955. The difference between the average score of both configurations is approximately 0.0043. This difference, albeit small, indicates that with a lower k value, the reputation scores of reliable sources are closer to the maximum possible score of 1.

The main reason behind the small difference in this scenario resides in the fact that in both configurations, the sources receive a high number (around 30,000) of maximum (or near-maximum) ratings. As seen in Figure 13, when k is lower, the speed at which reputation scores increase is higher. However, with a sufficiently high number of

TABLE 9. Evolution of maximum reputation scores.

Ratings	$k = 10$	$k = 250$
500	0.9901	0.8333
1000	0.9950	0.9000
10000	0.9995	0.9878
30000	0.9998	0.9959

ratings (30,000 in this case), both configurations eventually converge to the theoretical maximum reputation score of 1. This behavior can be observed in Table 9, which presents the evolution of the reputation scores for different numbers of maximum ratings (*i.e.*, best possible rating). This table is an extension of Figure 13, and supports the previous analysis by demonstrating that as the number of ratings increases, the reputation scores for both k evolve at different speeds, eventually converging to 1. In the case of the analysis presented in Table 8, if the dataset was to be reduced to only 500 or 1000 claims (instead of 30,000), the difference between both configurations would be more pronounced.

At the 80% noise level, the difference between the two configurations is slightly higher, at approximately 0.0112. This can be attributed to the fact that, in this high-noise scenario, the sources receive a more dispersed set of ratings (fewer maximum or near-maximum ratings), which affects the rate at which reputation scores evolve. Therefore, this difference reinforces the previous findings regarding the impact of k on the evolution of reputation scores.

These analyses demonstrate Nexus's sensitivity to changes in its key parameters. Adjusting γ enables fine-tuning Nexus's accuracy, while modifying ρ serves the same purpose by determining the influence between related claims. Adjusting k helps specify the evolution of source reputation scores, albeit at the expense of significantly increased execution time for larger datasets. This showcases the flexibility of Nexus, enabling the optimization of the proposed solution to the specific characteristics and demands of different non-authoritative environments.

1) COMPARISON WITH ESTABLISHED TRUTH DISCOVERY ALGORITHMS

A comparison of Nexus with TruthFinder and CRH using the synthetic datasets is summarized in Table 10. The results reported in this table represent the average of five executions conducted on the synthetic datasets described in Table 3, using the same noise levels, number of claims, and number of sources. For TruthFinder, γ was set to 0.15 and ρ to 0.9 to ensure a comparison with Nexus under similar conditions, as these are the same values as those used for Nexus. For CRH, the maximum number of iterations *iter* was set to 4 and the threshold t to 10^{-6} , as these values allow CRH to converge within a reasonable number of iterations without incurring excessive computational cost.

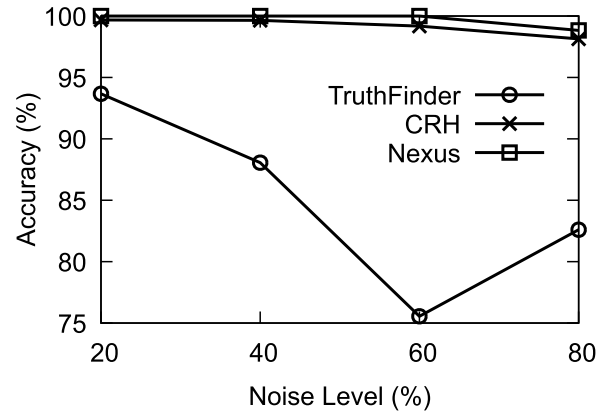
TABLE 10. Average similarity comparison for the synthetic datasets between TruthFinder, CRH, and Nexus.

Algorithm	Parameters	Dataset			
		synth-cat	synth-str	synth-con	synth-het
TruthFinder	$\gamma = 0.15$ $\rho = 0.9$	0.6	0.884	0.967	0.864
CRH	$iter = 4$ $t = 10^{-6}$	1.0	1.0	0.983	0.993
Nexus	$k = 10$ $\gamma = 0.15$ $\rho = 0.9$ $\lambda = 1.0$	1.0	1.0	0.996	0.999

For all assessed datasets, Nexus either outperforms or matches the performance of the remaining algorithms. For categorical and string data, both Nexus and CRH achieve maximum accuracy, while TruthFinder's performance is significantly lower (between 0.6 and 0.884). For continuous data, Nexus outperforms both TruthFinder and CRH, achieving an average similarity of 0.996 compared to 0.967 and 0.983, respectively. Lastly, for the heterogeneous dataset, Nexus achieves an average similarity of 0.999, outperforming both TruthFinder (0.864) and CRH (0.993).

The results presented in Table 10 demonstrate that Nexus outperforms or matches both TruthFinder and CRH across all data types. Compared to TruthFinder, and given that both algorithms were evaluated under identical γ and ρ configurations, the improvement achieved by Nexus is directly attributable to its reputation mechanism, which leverages historical source reliability to guide the truth discovery process. When compared to CRH, Nexus achieves higher accuracy for continuous and, consequently, heterogeneous data, while CRH presents a competitive performance for categorical and string data. This distinction stems from a fundamental difference in output strategy, whereby Nexus selects the candidate value with the highest confidence score, whereas CRH computes a weighted median using the normalized absolute deviation as the loss function. For instance, when estimating the age of an individual, Nexus would output a discrete value such as 18, 19, or 20, whereas CRH may return a non-integer result such as 19.25 as a consequence of the weighted median computation. Furthermore, although the normalized absolute deviation employed by the CRH implementation of Li [32] is a more robust estimator than the mean absolute deviation, it remains susceptible to outliers under high noise conditions. A substantial proportion of noisy claims can bias the weighted median toward values that diverge from the ground truth, a limitation that Nexus avoids by relying on confidence-score-based selection rather than a statistical aggregation function.

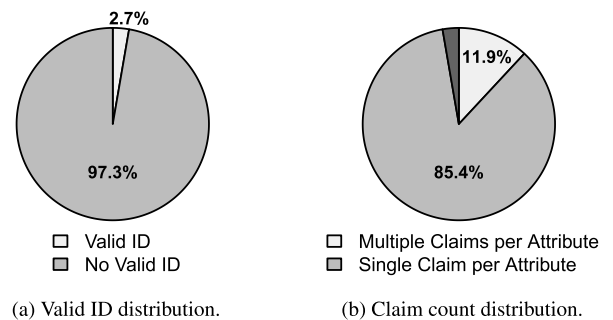
Overall, the results presented in this section demonstrate that Nexus is a capable solution compared to established

**FIGURE 14.** Average accuracy for different noise levels.

algorithms such as TruthFinder and CRH, highlighting its effectiveness and capability to accurately consolidate claims from multiple sources, even in scenarios with higher noise levels (60% and 80%), as shown in Figure 14.

B. SCENARIO 2: REAL-WORLD APPLICABILITY

This evaluation focuses on the applicability of Nexus in a real-world scenario, specifically in an environment where individuals lack official identification documents. Of the 5,444 unique individuals in the dataset, 5,296 did not provide a valid identification document number during the call, accounting for approximately 97.28% of the patients. This highlights the challenges LICs face in identity management, as the vast majority of individuals in this dataset lack official identification.

**FIGURE 15.** ID validity and claim count distribution.

Among individuals without valid identification, 649 (about 11.92%) possess at least one attribute with more than one provided claim. This indicates that these patients contacted the health line multiple times, either providing supporting or conflicting information for this attribute. On average, these individuals have 2.3 claims per attribute. The distribution of individuals with valid IDs and the distribution of claim counts per attribute for individuals without valid IDs are presented in Figures 15a and 15b, respectively. The percentages presented in Figure 15b correspond to the proportion of the total of individuals in the dataset (5,444).

TABLE 11. Reputation scores for best and worst sources.

	Source ID	Reputation Score
Best	22	0.760
	13	0.758
	10	0.754
Worst	15	0.550
	7	0.534
	14	0.527

Nexus successfully consolidated the claims associated with all 649 individuals who lacked valid identification and had multiple claims for at least one attribute (Figure 15b), producing a single unified record for each patient. The mean confidence score for the claims identified as most likely to be correct is approximately 0.687. Given that the average number of claims per attribute for these individuals is 2.3, this mean confidence score may be regarded as adequate for the context, as it is neither excessively high (*i.e.*, close to 0.9 or 1.0) nor excessively low (*i.e.*, close to the initial value of 0.5).

This process also enabled the identification of the most and least reliable sources (line operators) based on their reputation scores after consolidating all claims. Table 11 reports the three sources with the highest and lowest reputation scores following the processing of all claims in the dataset. Reputation scores range from 0 to 1, with all sources initially assigned a base score of 0.5, and higher values indicating greater source reliability. In this case, a clear distinction can be observed between the most and least reliable sources, with the best sources achieving reputation scores above 0.75, while the worst sources have scores below 0.55. This differentiation highlights Nexus's effectiveness in assessing source reliability, which can then be used in future interactions to weigh each source's claims accordingly.

The results derived from this real-world dataset demonstrate the practical applicability of Nexus in environments suitable for non-authoritative approaches. By effectively consolidating claims from multiple sources for individuals without valid identification, Nexus provides a mechanism through which the consolidated record may, for example, be used in conjunction with existing authoritative sources to associate it with existing information (when available). In doing so, Nexus facilitates the linkage of individuals to existing records as well as the enrichment of those records with additional or more up-to-date information.

Furthermore, in situations where authoritative information is unavailable, the consolidated records produced by Nexus can serve as a basis or foundation for establishing digital identities. These may subsequently be verified and formalized through official channels, thereby contributing to improved identity management in such challenging contexts. In addition to supporting the enhancement or creation of digital identities, the consolidated information concerning

individuals can also be utilized to improve service delivery, resource allocation, and policy-making across multiple sectors, including healthcare, social services, and humanitarian aid.

VI. DISCUSSION AND CONCLUSION

The previous sections presented the architecture and an evaluation of Nexus, focusing on its accuracy and sensitivity to key parameters. The results demonstrate that Nexus is a robust and effective solution, achieving high accuracy across both synthetic and curated datasets, even in scenarios with high noise. Moreover, Nexus allows fine-tuning several parameters that directly affect its performance and behavior, enabling it to adapt to different scenarios and requirements.

The real-world applicability of Nexus was also explored and demonstrated through an experiment using data from Guinea-Bissau's 24-hour Health Line. The results from this experiment highlight the potential of Nexus to address identity management challenges in non-authoritative environments, particularly in contexts where individuals lack official identification documents. By consolidating claims from multiple sources, Nexus can facilitate the creation of accurate and trustworthy identity records and improve service delivery in such settings. The flexibility and adaptability of Nexus make it particularly suitable for deployment in non-authoritative environments, where data quality and source reliability can vary significantly. The ability to adjust parameters such as the dampening factor, influence of related claims, and reputation system granularity allows Nexus to be tailored for specific use cases and adversarial settings, ensuring optimal performance even in challenging conditions (such as high noise or low variability).

However, before deploying Nexus in a production environment, it is crucial to conduct a thorough analysis of data characteristics, expected noise levels, and potential adversarial behaviors. This initial assessment is one of the most important steps to ensure the successful application of Nexus in any real-world scenario. As demonstrated in the evaluation, data characteristics, such as low variability in categorical data, can significantly impact the accuracy of the truth discovery process, further reinforcing the need for a comprehensive understanding of the data being provided to the system prior to deployment.

Regarding noise levels, this work presented an analysis of levels up to 80% with known noise characteristics (*e.g.*, number of noisy sources). While Nexus demonstrated strong performance even in these high-noise scenarios, it is important to understand that in real-world applications, noise levels are not known a-priori, whilst varying significantly and unpredictably. This challenges the assessment of whether the output of the consolidation step accurately depicts reality.

The unpredictable nature of both noise characteristics and source behavior emphasizes the significance of conducting a thorough adversarial analysis. Distinguishing between random noise and targeted adversarial actions is crucial to ensuring the integrity of the truth discovery process.

This analysis should consider various adversarial strategies, such as collusion among sources [64], [65], [66], Sybil attacks [67], [68], [69], and data poisoning [70], [71], [72]. Without a proper adversarial analysis, deploying Nexus in a production environment may lead to suboptimal results, as the system may be vulnerable to manipulation by malicious sources. Given that the goal of Nexus is to facilitate the construction of trustworthy identities in non-authoritative environments, which ultimately may be utilized within authoritative or official systems, it is essential to understand potential adversarial strategies and to implement appropriate countermeasures.

Beyond technical considerations, the deployment of Nexus raises important data protection and ethical implications that must be addressed prior to operation. As a system that processes sensitive personal data originating from non-authoritative sources, Nexus constitutes a personal data processing activity under applicable regulatory frameworks. The GDPR requires that such processing rest on a lawful basis, such as the performance of a task in the public interest, and comply with the principles of purpose limitation and data minimization. In LIC and LMIC deployments, where national data protection legislation may be absent or weakly enforced, these principles nonetheless provide a normative baseline consistent with the SDG 16.9 objectives that motivate this work. To mitigate the risks associated with processing data from vulnerable populations, such as undocumented individuals in LICs, deployments of Nexus must implement explicit data-use policies, access-control mechanisms, and informed-consent protocols, ensuring that individuals whose data is consolidated retain meaningful agency over its use and scope.

Even considering these general safeguards, two specific misuse risks warrant acknowledgment. First, reputation scores could be repurposed as discriminatory signals, for instance, to systematically discount claims from sources associated with particular communities. Thus, governance policies should therefore restrict their use to the truth discovery process itself. Second, an incorrectly consolidated record could deny an individual access to services or introduce attribute conflicts with serious consequences in healthcare or legal contexts. While Design Goal 5 addresses source-level errors, individual-level correction remains an open challenge. Future work should define a correction protocol that flags disputed attributes for re-evaluation, with the confidence score serving as a natural trigger so that attributes below a configurable threshold can automatically route to human oversight before use in downstream authoritative processes.

Regarding this aspect, future work will focus on augmenting the adversarial analysis presented in Section III-E, exploring additional attack vectors and strategies that may be employed by malicious sources to manipulate the truth discovery process. An extension to this analysis will include the evaluation of additional data types (instead of focusing solely on categorical data) and the exploration of more

complex scenarios. This will provide additional inequalities such as the one presented in (13), which will help define safe operational boundaries for Nexus' parameters in adversarial settings.

Another important aspect for future work resides in updating the reputation evolution within the reputation system. Currently, the reputation scores of sources increase and decrease at the same rate (as seen in Figure 13). However, in real-world scenarios, a more realistic approach would involve implementing asymmetric reputation evolution, where sources lose reputation more rapidly than they gain it. This approach reflects the common real-world phenomenon where trust is harder to build than to lose. Prior work on trust dynamics suggests that trust and reputation often evolve asymmetrically, with negative events exerting a disproportionate impact on future trust assessments, while positive behavior typically accumulates more gradually [73], [74], [75]. In evolutionary trust-game settings, asymmetric parameters can substantially alter the resulting trust dynamics, reinforcing the idea that symmetric update rules may fail to capture realistic credibility evolution [73]. This behavior has the potential to improve the overall accuracy of the truth discovery process, whilst also discouraging sources from providing inaccurate data, as the repercussions on their reputation would be more severe.

Lastly, future work will also focus on the implementation of a duplicate detection mechanism. In real-world scenarios, it is common for multiple representations of the same individual to exist within a system [49], [57], [76], [77]. In an identity management context, this can lead to fragmentation of an individual's identity across multiple records. Correctly identifying and merging these duplicate records is crucial to ensuring the integrity and reliability of identity records within Nexus. An initial step of such a mechanism is already performed, as the similarity functions used in the truth discovery process can also be employed to identify potential duplicate records. However, further research is required to develop a comprehensive duplicate detection and merging strategy capable of handling the complexities associated with real-world digital identity data. In addition to technical improvements, future work should also investigate integrating consent management frameworks, such as user-controlled data access tokens or selective disclosure mechanisms compatible with the SSI model, to ensure that individuals whose data is consolidated retain agency over how their identity records are used.

This work presented Nexus, a novel data fusion module designed to enable the construction of trustworthy digital identities in non-authoritative environments. By combining truth discovery algorithms with a robust reputation system, Nexus effectively consolidates claims from multiple sources, even in scenarios characterized by high levels of noise and potential adversarial actors. Several experiments were conducted using both synthetic and curated datasets, demonstrating Nexus's ability to achieve high accuracy across various

scenarios. Furthermore, the system's flexibility allows for fine-tuning of key parameters, supporting its adaptation to different use cases and requirements. At this stage, Nexus offers a practical approach to non-authoritative identity, with promising results, and represents a solid step towards an inclusive identity solution.

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